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SQL Practical Exercise

Bright Light Tutorials.

1 SQL Joins

1. Inner Join

```
SELECT e.employeeID, e.FirstName, e.Lastname, e.Department,
       e.salary, p.Project ID, p.ProjectName, p.Budget,
       p.Status
FROM employees.e
INNER JOIN projects.p
    ON e.employeeID = p.employeeID;
```

employee ID	First Name	Last Name	Department	Salary	project ID	Project Name	Budget	Status
1	John	Doe	IT	70 000	101	AI Developm	100 000	Completed
1	John	Doe	IT	70 000	103	Cybersecurity Audit	75 000	Pending
2	Alice	Smith	HR	60 000	102	Employee Training	50 000	Ongoing
3	Bob	Johnson	Finance	75 000	104	Financial Analysis	90000	Ongoing
5	Emma	Wilson	Sales	72000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80 000	106	Risk Management	80000	Pending

1. **Full outer Join :**

2. Left Join

```
SELECT e.employeeID, e.FirstName, e.LastName, e.Department, e.Salary,
       p.ProjectID, p.ProjectName, p.Budget, p.Status
FROM employees.e
LEFT JOIN projects.p
ON e.employeeID = p.employeeID;
```

EID	First Name	last Name	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70 000	101	AI Development	100 000	Completed
1	John	Doe	IT	70 000	103	CyberSecurity Audit	75 000	Pending
2	Alice	Smith	HR	60 000	102	Employee Traing	50 000	Ongoing
3	Bob	Johnson	Finance	75 000	104	Financial Analysis	90 000	Ongoing
4	David	Brown	IT	72 000	Null	Null	Null	Null
5	Emma	Wilson	Sales	65 000	105	Market Expansion	65 000	Completed
6	Michael	Clark	finance	80 000	106	Risk Management	80 000	Pending

3. Right Join

```
SELECT p.ProjectID, p.ProjectName, p.Budget, p.Status, e.employeeID,
       e.FirstName, e.lastName, e.Department, e.Salary
FROM employees.e
RIGHT JOIN project.p
ON e.employeeID = p.employeeID;
```

Project ID	First Name	last Name	Department	Salary	Employee ID	Project Name	Budget	Status
101	John	Doe	IT	70 000	1	AI Development	100 000	Completed
102	Alice	Smith	HR	60 000	2	Employee Training	50 000	Ongoing
103	John	Doe	IT	70 000	1	Cybersecurity Audit	75 000	Pending
104	Bob	Johnson	finance	75 000	3	Financial Analysis	90 000	~~Pending~~
105	Emma	Wilson	Sales	65000	5	Market Expansion	65 000	Ongoing
106	Michael	Clark	finance	80 000	6	Risk Management	80 000	Completed Pending

4. ## Full outer Join :

```
SELECT e.employeeID, e.FirstName, e.LastName, e.Department,
       e.Salary, p.projectID, p.ProjectName, p.Budget,
       p.Status
FROM e.employees
FULL OUTER JOIN p.project
   ON e.employeeID = p.employeeID;
```

Employee ID	First Name	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70 000	101	AI Development	100 000	Completed.
1	John	Doe	IT	70 000	103	~~AI Development~~ Cybersecurity Audi	75 000	Pending
2	Alice	Smith	HR	60 000	102	Employee Traing	50 000	Ongoing
3	Bob	Johnson	Finance	75 000	104	Financial Analysis	90 000	Ongoing
4	David	Brown	IT	72 000	Null	Null	Null	Null
5	Emma	Wilson	Sales	65 000	105	Market Expansion	65 000	Completed
6	Michael	Clark	Finance	80 000	106	Risk Management	80 000	Pending.

2. ## Unions & Union All

5. ## Alias

```
SELECT DISTINCT City AS Location
FROM employees
UNION
SELECT DISTINCT project_Status AS Location
FROM projects;
```

Location
New York
Los Angels
Toronto
London
Sydney
Completed
Ongoing
Pending

```
6. SELECT   city AS Location
   FROM     employees
   UNION    ALL
   SELECT   project_status AS location
   FROM     projects;
```

Results:

Location
New York
Los Angels
Toronto
London
Sydney
New York
Completed
Ongoing
Pending
Ongoing
Completed
Pending

3. Filtering Statements:

```
7. SELECT   employee_ID, firstName, LastName, Department,
            salary,
   FROM     employees,
   WHERE    salary > 70 000;
```

employee ID	first Name	LastName	Department	Salary
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000
6	Michael	Clark	Finance	80 000

Filtering Statements

8.

```
SELECT employeeID, firstName, LastName, Department,
       salary,
  FROM employees,
  WHERE department = 'IT' AND department = 'finance';
```

employee ID	firstName	LastName	Department	Salary
1	John	Doe	IT	70 000
3	Bob	Johnson	finance	75 000
4	David	Brown	IT	72 000
6	Michael	clark	Finance	80 000

9.

```
SELECT projectID, projectName, Budget, Status
  from projects
  WHERE Status < > 'Completed';
```

Project ID	~~pro~~ Budget	project Name	Status.
102	50 000	Employee Training	Ongoing
103	75 000	Cybersecurity Audit	Pending
104	90 000	Financial Analysis	Ongoing
106	80 000	Risk Management	Pending.

Filtering Statements :

10.
```
SELECT projectID, projectName, Budget, status
FROM projects
WHERE Budget > 70 000 AND Status <> 'Completed';
```

projectID	project Name	Budget	Status
103	Cybersecurity Audit	75 000	Pending
104	Financial Analysis	90 000	Ongoing
106	Risk Management	80 000	Pending

11.
```
SELECT employeeID, FirstName, LastName, Department,
       Salary, City
FROM employees
WHERE city = 'New York' AND city = 'Toranto'
ORDER BY salary DESC ;
```

EmployeeID	FirstName	LastName	Department	Salary	City
6	Michael	Clark	Finance	80 000	New York
4	David	Brown	Finance	75 000	Toranto
1	John	Doe	IT	70 000	New York

12.
```
SELECT employeeID, firstName, LastName, Department, Salary
FROM employees
ORDER BY Salary DESC
LIMIT 3 ;
```

EmployeeID	FirstName	LastName	Department	Salary	
6	Michael	Clark	Finance	80 000	
3	Bob	Johnson	Finance	75 000	
4	David	Brown	IT	72 000	

Aggregate functions: with GROUPBY & Having:

13. Total salary by department.

```
SELECT Department, sum(Salary) AS Total Salary
FROM employees
GROUP BY department
ORDER BY sum(salary) DESC;
```

Department	Total Salary
Finance	155 000
IT	142 000
Sales	65 000
HR	60 000

14. Avg Salary per city where salary is > than 65 000

```
SELECT City, AVG(salary) AS Average Salary
FROM employees
GROUP BY city
HAVING AVG(salary) > 65 000;
```

City	Average Salary
London	72 000
New York	75 000
Toranto	75 000

Aggregate functions.

15. Number of employees per department

```
SELECT Department , COUNT(employee ID) AS Employee cou
FROM employees
GROUP BY Department
HAVIGING COUNT(employeeID) > 1;
```

Department	Employee count
Finance	2
IT	2

16 Number of projects per Status where atleast 2 projects.

```
SELECT Status , count(projectID) AS Project Count
FROM projects
GROUP BY Status
HAVING count(project ID) >= 2;
```

Status	Project Count
Completed	2
Ongoing	2
Pending	2

17. Total budget per employee where budget is > 150 000

```
SELECT e.employeeID, e.FirstName, e.LastName,
       sum(Budget) AS 'Total project budget'
FROM employees
JOIN projects
ON e.employeeID = p.employeeID
GROUP BY e.employeeID
HAVING sum(p.Budget) > 150000;
```

Employee ID	FirstName	LastName	Total project Budget
1	John	Doe	175 000